HOW TO MANAGE SOFTWARE DEVELOPMENT BY EMPANELLED AGENCIES

REVISION HISTORY

Revision No./Version No.	Date	Prepared by	Approved by	Details of Change
1.0	10-Jan-2024	Prachi Naik (Asst. Manager IT)	Anant Sham Yende (CTO)	
2.0	15-Jul-2024	Prachi Naik (Asst. Manager IT)	Anant Sham Yende (CTO)	Changes in Annexure Table

OBJECTIVE

To ensure the empanelled agency involved in development of application and database develop and implement software under technical standards of GEL and latest complete code and database is in custody of GEL at any given time and GEL team is well aware about the technicalities of the projects.

PARTICIPANTS

1. Empanelled Agency(EA): The empanelled agency who will be developing the software..

2. GEL Team: The Project Manager(PM) and Technical Lead(TL) assigned for the project.

PROCEDURE TO FOLLOW

Project Initiation

1. For each project a TL and PM will be assigned from GEL & the EA side.
2. PM[EA] to prepare a project plan, including timelines, milestones, and deliverable as per the scope of work shared by GEL. This plan to be approved by PM [GEL].
3. TL [GEL] to decide on the technology in which the software will be developed ensuring preference is given to open source. Inputs on best feasibility & list of technology relevant to be given by TL [EA].

4. PM[EA] to submit details of project team with appropriate roles and responsibilities to the PM[GEL].

5. TL [GEL] will coordinate with the TL[EA] and list requirement of source code repositories (Gitlab) for the project. After that create repositories on Gitlab and share with TL[EA] & PM[EA] and add EA team members for applicable repositories.

6. PM [GEL] will coordinate with team and provide credentials for accessing GEL JIRA mainly to manage the issues and tracking of bugs.

Analysis and Design

1. PM[EA] & TL[EA] has to prepare and submit all the documents to GEL as per Annexure 1 at various stages.

2. Every Meeting between team and client to be minuted. PM [GEL] to ensure every document including all MOMs is uploaded on GEL TM at various phases.

Development

1. TL[EA] will set up a development environment and link with Gitlab shared by GEL. TL[EA] has to mandatory make use of Gitlab repositories to maintain code and database scripts and versioning of same.

2. TL[EA] has to follow coding standards and best practices for efficient and maintainable code.

3. TL[EA] has to develop and implement the software components according to the specifications. Software Technical design(specifications) as to be submitted as mentioned in Annexure 1.

4. Any external applications/plugins/licensing libraries to be used by TL[EA] to be approved by GEL team strictly on mail.

Testing

TL[EA] & PM[EA] has to ensure testing of application before roll out and share the complete progress report with GEL. Testing will include the following:

1. Creating test cases and test scenarios based on requirements. The test scenarios cases should be shared with TL[GEL].

2. Performing unit testing to ensure individual components work correctly.
3. Conducting integration testing to verify the interaction between different modules.
4. Executing system testing to validate the software against the specified requirements.
5. Performing performance, security, and usability testing as necessary.
6. Third party security audit has to be also conducted before moving to production and once every year and all the vulnerabilities has to be fixed.

Deployment, Release & Hosting

1. The production environment will be either procured by GEL or EA as per the scope. The access to the production server will be shared by PM[GEL].
2. For government projects the sites will be primarily hosted at NIC State Data Center (Meghraj). If government decides to host on any private cloud the same needs to be as per the MEITY guidelines and only on Cloud Service Providers(CSP) empanelled with MEITY.
3. SSL and Domain name will be provided by GEL and charges will be as per scope of work.
4. TL[EA] has to ensure that deployment planning and environment setup is done and the same has to be communicated with the TL[GEL].
5. TL[EA] has to perform deployment activities, including installation, configuration, and data migration.
6. TL[EA] has to prepare the software for release, including versioning and release notes. EA team will discuss and finalize the deployment times in coordination with TL[GEL] to have least downtime in production hours.

Maintenance and Support

1. In case any changes being done in the software application based on requirements or inputs PM[EA] to ensure change request SOP of GEL is followed. TL[EA] to ensure no changes are done without CR.
2. EA has to provide post-release support for bug fixes, enhancements, and troubleshooting and ensure issue tracking, prioritization, and resolution towards the issues reported and handled

through GEL JIRA strictly.

3. PM[EA] & TL[EA]has continuously monitor and optimize the software performance and security. PM[GEL] to review progress of the projects with PM[EA] on a periodic basis as a part of plan.

Project Closure

1. PM[EA] has to conduct a project review to evaluate the success criteria and lessons learned.
2. PM[EA] has to document the project closure report, including achievements and challenges.
3. PM[EA] has to handover project artifacts, code, and documentation to GEL for future reference.

Documentation

1. PM[EA] has to prepare and submit all documents as mentioned at Annexure 1 as per the standard format shared by GEL at the respective phase of the project.
2. PM[EA] has to maintain and submit user manuals and up-to-date documentation throughout the development process. PM has to ensure it is uploaded in GEL TM.
3. PM[EA] has to maintain Change request, client meeting minutes, training reports and feedback. All documents to be shared with PM[GEL].

Annexure 1

Sr No	Document	Phase of Submission
1	Functional Requirement specification(FRS) as per GEL format	Design & Analysis
2	Technical Design Document will comprise of following	Design & Analysis
a	Software architecture, Development architecture, Deployment architecture	Design & Analysis
b	Technical specifications (including technology to be used with version details for application/db/reports etc)	Design & Analysis
c	Database schema with description of each table and fields and relationships.	Design & Analysis
d	User interface design document.	Design & Analysis
e	Functional specifications(which will give detailed functionality of all components.	Design & Analysis
3	Test Plan and Test scenarios	Design & Analysis
4	Sprint Plan	Development
5	Project artifacts	Project closure
6	Project Code	Should be committed in GIT lab on real time basis and handed over during project closure.
7	Project User manuals	During any module Delivery or project closure
8	Requirement Traceability Matrix(RTM) of the Project as per GEL format	Project closure or as per request of GEL